



Yiannis Dimitrakopoulos

Software Developer

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Profile

Software engineering student at Codam College (42 Network) building toward AI/ML engineering roles. Hands-on experience implementing constrained decoding, hybrid retrieval, and graph-based systems for LLMs from first principles — working directly with raw model internals rather than high-level frameworks.

Languages

English

Proficient in English (Certificate of Proficiency in English – ECPE)

Skills

Programming Languages

Python, C, JavaScript

AI / ML & LLM Systems

Constrained decoding (FSM, logit masking), hybrid retrieval (BM25 + dense, RRF fusion), knowledge graphs (Neo4j), open-source LLM inference (Qwen3)

Front-End Development

Vue.js, Nuxt Js, Tailwind CSS

Back-End Development

Node.js, Express.js

Version Control

Git/GitHub, Bash (Unix Shell)

Projects

Call Me Maybe

Constrained Decoding Engine for LLMs
2026

- Built a full constrained-decoding pipeline from scratch for Qwen3-0.6B — manual logit masking, an FSM tracking valid JSON states, and a generation loop on raw logits, no ML frameworks. Implements the mechanism used inside vLLM/SGLang for structured output.

RAG Against The Machine

Hybrid Retrieval System
2026

- Built a hybrid BM25 + dense retrieval pipeline with Reciprocal Rank Fusion (RRF), a caching layer, and two chunking strategies — AST-based for Python source, plain chunking for markdown/text.

Constrained-GraphRAG

Constrained-Decoding GraphRAG
In progress, 2026

- Composing the retrieval system above with FSM-based constrained decoding to extract entities and relationships into a Neo4j knowledge graph, enabling multi-hop graph-augmented retrieval across code and documentation.

Lely Hackathon — Embedded Programming, 11/2025

- Programmed mBot robots via Python in a 1-day embedded-systems challenge under time pressure.

Build Weekend Amsterdam — Young Creators, 11/2025

- Built “**ema.**”, a voice assistant guiding older adults through daily travel using n8n, ElevenLabs, and Twilio. **Finalist** among multiple teams.

Education

Codam College

Amsterdam

10/2025 – Present

- Intensive, project-based curriculum focused on **C, Python, algorithms, and collaborative software engineering.**
- Emphasis on **teamwork, problem-solving, and hands-on projects** to build real-world software skills.

Aristotle University of Thessaloniki (AUTH)

Bachelor's Degree, Physical Education & Sport Science

10/2017 – 07/2021

Professional Experience

Front Desk Staff

Hertz Autohellas

04/2024 – 10/2024

- Customer Service Skills:** Provide customer service, addressing customer inquiries, concerns, and requests.
- Communication Skills:** Effectively communicating with customers and colleagues.
- Attention to Detail:** Ensuring accuracy in all tasks.
- Teamwork:** Collaborated with colleagues positively.

Greek Army

Secretary Assistant

- Served an obligatory twelve-month period, as a secretary assistant and assistant for validation of expenditures.
- Developed interdisciplinary collaboration and adaptation.
- Learned to coexist and cooperate with a squad of people.
- Gained independence by managing personal responsibilities and adapting to military life.

Courses

Developing Back-End Apps with Node.js and Express

IBM

Introduction to Containers w/ Docker, Kubernetes & OpenShift

IBM

Programming with JavaScript

META

Version Control

META

Foundational C# with Microsoft

freeCodeCamp